

Git vs Other VCS

Feature	Git (Distributed)	SVN / CVS (Centralized)	Mercurial	Perforce
Model	Distributed	Centralized	Distributed	Centralized
Offline Work	✅ Full functionality offline	❌ Requires network to commit/update	✅ Yes	❌ Limited
Repository Copy	Each user has full repo	Only central repo exists	Each user has full repo	Only server stores full history
Performance	Very fast (local operations)	Slower (depends on server)	Similar to Git	Fast for large binary files
Branching/Merging	Easy, fast, and cheap	Slower and more error-prone	Similar to Git	Basic branching support
Popularity	Most popular globally	Older, legacy systems	Less popular than Git	Used in enterprise, game dev
Ease of Use	Moderate learning curve	Simpler for beginners	Simpler CLI than Git	Complex but powerful
Conflict Handling	Manual resolution when needed	More frequent conflicts	Similar to Git	Automatic + manual merge tools
Use Cases	Open source, enterprise, solo devs	Legacy enterprise projects	Simpler DVCS projects	Game assets, large-scale enterprise
Tooling & Integrations	Extensive (GitHub, GitLab, CI/CD)	Good but outdated	Moderate	Enterprise-grade, limited open tools
Security	SHA-1 hash-based, good history integrity	Weaker integrity controls	Similar to Git	Strong enterprise access control